Information Overload under Rational Learning*

Jonghyun Park† Sudipta Sarangi‡ Hector Tzavellas§

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Abstract

In canonical models of learning, additional information is always beneficial because agents can freely ignore signals. We depart from this framework by introducing cognitive costs that scale with the dimensionality of the information environment. An agent chooses an inference rule to trade off accuracy against these costs. We characterize conditions under which information overload arises and show that, as dimensionality increases, optimal behavior attenuates responsiveness and can collapse to a prior-based rule. Under Gaussian assumptions, the optimal rule is affine and features a cognitive deflator that generates the familiar inverted-U relationship between information load and accuracy documented in the experimental literature on overload. The model also provides a rational foundation for behavior resembling correlation neglect and yields implications for social learning, including the existence of an optimal degree that balances accuracy gains against cognitive costs.

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†Department of Economics, Virginia Tech. E-mail: jongh2419@vt.edu

‡Department of Economics, Virginia Tech. E-mail: ssarangi@vt.edu

§Department of Economics, Virginia Tech. E-mail: hectortz@vt.edu

1 Introduction

Decision-makers increasingly operate in environments characterized by an abundance of information. Improvements in data collection, communication technologies, and social connectivity have increased the number of signals agents can observe and employ towards decision tasks. At the same time, a broad empirical literature shows that decision quality does not necessarily improve with greater information availability and may instead deteriorate when individuals are exposed to excessively large amounts of information (Eppler and Mengis, 2004; Roetzel, 2019).[1] This phenomenon, commonly referred to as *information overload*, raises a natural question for economic theory: *how can rational agents perform worse when they have access to more information?*

Canonical economic models of learning and selective information usage do not provide an answer to this question. In standard Bayesian decision theory (Blackwell, 1953), rational inattention (Matějka and McKay, 2015; Sims, 2003), models of costly information acquisition (Grossman and Stiglitz, 1980; Vives, 1993), or sparsity-based approaches (Gabaix, 2014), additional information is always (weakly) beneficial. The reason is that these frameworks satisfy a form of free disposal of information: agents can choose how much information to use and can disregard signals that are not processed at no cost. Consequently, additional information cannot reduce performance, since it expands feasible learning strategies without increasing implementation costs.

A feature missing from these approaches is processing costs that *increase* with the dimensionality of the information environment. Experimental evidence shows that when exposed to an increased number of cues, decision-makers adjust their search behavior, inspect a smaller share of available information, and rely more heavily on screening (Payne, 1976). Moreover, even information that is ultimately irrelevant to the underlying decision task has been shown to affect decision time and increase error rates (Chadd et al., 2021; Dummel et al., 2016). These findings indicate that the *presence* of additional signals is *itself* costly since implementing a learning strategy requires screening and filtering. Richer information environments can thus make learning rules more costly to implement, providing a candidate mechanism through which additional information can reduce performance.

Motivated by this evidence, we model information overload as a property of optimal inference

[1]For instance, experiments in consumer choice document poorer decisions in the form of lower choice quality or fewer correct selections (Jacoby et al., 1974; Malhotra, 1982). In accounting contexts, studies find that greater information can reduce judgment quality, as reflected in lower consistency and weaker agreement with benchmark evaluators (Chewning Jr and Harrell, 1990). In social media settings, as individuals are exposed to more content, their responsiveness declines and any given exposure becomes less effective toward adoption (Rodriguez et al., 2014).

under the presence of *dimension dependent costs*. We consider an agent who observes a set of $(d+1)$ signals and chooses a learning rule ℓ to estimate an unknown state. The agent trades off statistical accuracy, as measured by the mean squared error, against a cognitive cost that depends both on the responsiveness of the chosen rule (its induced variance), and on the dimensionality of the information environment. This formulation captures the idea that the presence of additional signals can itself be costly, so that even simple or low-dimensional rules may become expensive to implement in sufficiently information rich environments.

This perspective explains how more information can lower performance even when agents behave fully rationally. While agents can always choose to ignore additional signals, doing so is not the puzzle; rather, the question is why ignoring information can be optimal even when it would improve statistical accuracy. In our framework, each additional signal expands the set of feasible learning rules but also increases the cost of implementing any rule that remains responsive to the data. When cognitive costs scale with the dimensionality of the information environment, higher-dimensional settings make responsiveness increasingly expensive. As a result, the agent optimally attenuates responsiveness by placing less weight on observed signals and shifting weight toward prior information. In the limit, inference collapses: the optimal rule converges to a constant, and learning accuracy returns to the prior variance.

Our first contribution is to characterize this mechanism at a general level. We provide necessary and sufficient conditions under which information overload arises in high-dimensional environments and show that it takes a stark form. As the dimensionality of the information environment increases, the mean-squared error under the optimal learning rule converges to the prior benchmark. This result does not require a specific parametric form of the cost function or on congestion in attention or communication. Instead, it follows from two properties: (i) constant rules are costless, so the agent can always revert to a prior-based estimate, and (ii) when cognitive costs scale with the dimensionality of the information environment, any sequence of learning rules that remains meaningfully responsive becomes prohibitively costly to implement. Under these conditions, sufficiently rich information environments lead to a complete collapse of inference to the prior.

Next, we characterize the structure of optimal learning. We show that when costs are increasing in the variance of learning rules, and the Bayesian posterior mean is affine in signals, the optimal learning rule is also affine in the signals. This result provides a structural foundation for the use of linear updating rules, such as DeGroot (1974) learning, in social networks. It demonstrates that these rules are not merely behavioral approximations, but can be understood as exact solutions to the trade-off between accuracy and cognitive burden.

The tractability of affine learning allows us to characterize how optimal behavior varies with the dimensionality of the information environment. Assuming Gaussian prior information, the optimal affine rule features a simple *cognitive deflator* that scales down the weights on all signals as d grows. As a result, learning performance is generically non-monotone in the dimension of the information environment. Specifically, the mean squared error initially falls as additional signals improve statistical accuracy, but eventually increases because expanding the dimension of the information environment raises cognitive load and induces shrinkage toward the prior. This pattern can be summarized as a *U-shaped* mean squared error as a function of information load (equivalently, an inverted-U in decision accuracy).

Finally, we present two applications of our framework. First, we apply the model to correlation neglect, the empirical regularity that agents treat redundant sources as if they were independent (e.g., Enke and Zimmermann, 2019). In our framework, correlation neglect-like behavior emerges not as an outcome of misspecified learning, but as a consequence of optimal shrinkage in high-dimensional environments. Specifically, we show that under the Gaussian-quadratic benchmark, the distance between a *correlation-aware* optimal rule that is fully rational and internalizes correlation structure, and a misspecified rule that neglects correlation, converges to zero as d grows. This occurs because, as informational dimensionality increases, the gains from adjusting decision weights to reflect covariance structure become negligible relative to the associated cognitive costs. Our model therefore predicts that behavior akin to correlation neglect is most likely to arise in high-dimensional settings with many information sources, consistent with the experimental findings of Dasaratha and He (2021).

Second, we explore a social learning interpretation of our model by mapping the information dimension d to an agent's degree in a network. Invoking our affine characterization results, an agent who observes her own signal and those of her d_i neighbors adopts a DeGroot-style aggregation rule with a cognitive deflator that uniformly shrinks responsiveness as degree rises. This implies that, even holding signal quality fixed, there is a finite degree that balances the marginal accuracy gain from additional neighbors against the marginal cognitive cost of processing their signals. When cognitive costs are sufficiently large relative to signal quality, additional links are strictly harmful; otherwise, losses are minimized at an interior degree that decreases with signal noise and increases with signal precision.

This perspective has two implications. First, if agents choose how many sources to follow based on these trade-offs, differences in cognitive capacity can generate skewed degree distributions, with many low-degree agents and a small number of highly connected hubs. Second, the model has implications for informational interventions in networks. Standard seeding strategies target highly connected agents (Banerjee et al., 2013), but in our setting, centrality does not necessarily make an agent an effective conduit for additional information. As informational dimension increases with degree, the cognitive deflator attenuates responsiveness, so high-degree agents place less weight on any given signal. As a result, additional information has a smaller marginal impact on their behavior, and targeting highly connected nodes need not maximize diffusion or behavioral change.

Related Literature

A classical benchmark in Bayesian decision theory is that more informative signal structures weakly improve statistical performance. Under quadratic loss, Bayes risk is non-increasing in the Blackwell order due to *free disposal*. That is, when additional information is available, it can be ignored at no cost, so the feasible set of decision rules is nested (Blackwell, 1953). Our framework preserves free disposal at the level of feasibility but breaks the monotonicity result by allowing the *implementation cost* of inference to scale with the dimensionality of the information environment. The key force is that richer environments make responsive inference more costly.

This perspective is also distinct from other strands of the learning literature such as rational inattention models, where agents choose information subject to a cost, typically based on mutual information (Matějka and McKay, 2015; Sims, 2003). These models generate selective attention, but additional *available* signals do not worsen learning accuracy, since ignoring information is costless. Related sparsity-based approaches penalize the use of many signals or large weights (Caplin and Dean, 2015; Gabaix, 2014). By contrast, our framework does not model endogenous acquisition. Instead, costs depend on the dimensionality of the information environment itself, so that even sparse or low-dimensional rules become less attractive in complex environments.

A large literature studies social learning and information aggregation, often taking linear or DeGroot-style updating as a primitive (Banerjee, 1992; DeGroot, 1974; Golub and Jackson, 2010; Golub and Sadler, 2016; Ottaviani and Sørensen, 2001). Molavi et al. (2018) provide behavioral foundations for such rules. By contrast, we derive affine updating as the optimal solution to a Bayesian inference problem with cognitive costs, providing a rationalization for these commonly used rules.

Our results are also related to the literature on choice overload. For instance Iyengar and Lepper (2000) and Besedeš et al. (2012) show that larger choice sets can reduce engagement and performance, often attributed to psychological mechanisms. Our framework provides a complementary rational explanation. Specifically, when additional options increase the dimensionality of the evaluation problem, they raise the cost of responsive inference, leading agents to attenuate responsiveness and rely more on coarse rules.

Our applications connect to the literature on correlation neglect and misspecified learning. Enke and Zimmermann (2019) document correlation neglect in belief formation, while Dasaratha and He (2021) show that sparse networks can outperform dense ones in learning environments. On the theory side, Dasaratha and He (2020), Esponda and Pouzo (2016), Frick et al. (2020), and Spiegler (2016) study inference under misspecification or naive reasoning. By contrast, agents in our model hold correct beliefs. Correlation neglect-like behavior arises because, in high-dimensional environments, optimal shrinkage makes the behavioral difference between correlation-aware and correlation-neglect rules small.

Finally, as mentioned above, a substantial empirical literature documents non-monotonic relationships between information quantity and decision performance, often finding inverted-U patterns (e.g. Chewning Jr and Harrell (1990)). Our contribution is to provide a formal mechanism that delivers this pattern under rational choice, to characterize general conditions under which overload arises, and to connect the mechanism to social learning and correlation neglect.

The rest of the paper is organized as follows. Section 2 presents the model. Section 3 characterizes information overload under rational learning. Section 4 derives the structure of optimal learning under information overload. Section 5 conducts comparative statics. Section 6 applies our results to social learning and correlation neglect. Section 7 concludes.

2 Model

This section presents a learning problem in which an agent faces an information environment of increasing dimensionality. Our modeling approach focuses on two distinct objects: (i) the *statistical environment*, which describes the information available at a given informational dimension, and (ii) the *processing technology*, which captures how costly it is to transform that information into an estimate. This separation helps us to define and analyze information overload without relying on a specific underlying distortion such as limited attention, rational inattention, or misspecification,all of which relate to processing technology, while still obtaining sharp implications.

Information Environment. There is an unknown state of the world $\theta \in \mathbb{R}$, drawn from a known prior distribution with finite second moment, $E[\theta^2] < \infty$. Let $\mu_\theta := E[\theta]$ and $\sigma_\theta^2 := \mathrm{Var}(\theta)$.

For each integer $d \geq 0$, the agent has access to an information environment consisting of

$d + 1$ signals,[2]

$$S^{(d)} := (s_0, s_1, \dots, s_d)^\top \in \mathbb{R}^{d+1}.$$

At this stage, we impose no parametric restrictions on the joint distribution of $(\theta, S^{(d)})$. The signals may be biased, correlated, heterogeneous in quality, or contain redundant components. We assume that all such features are contained in the joint law of $(\theta, S^{(d)})$. Throughout, expectations and variances are taken with respect to the joint distribution of $(\theta, S^{(d)})$.

Learning Rule. At any information level d, the agent can select a learning rule

$$\ell : \mathbb{R}^{d+1} \to \mathbb{R},$$

which maps the observed signal vector $S^{(d)}$ into an estimate $\ell(S^{(d)})$ of θ. We restrict attention to rules with finite second moment:

$$\mathcal{L}^{(d)} := \big\{\ell(S^{(d)}) \mid E[\ell(S^{(d)})^2] < \infty\big\}.$$

This restriction is standard in quadratic-loss environments and ensures that both mean squared error and the variance of the estimate are well-defined.

Consistency and Nested Environments. We impose a consistency condition across dimensions: when the environment expands (d increases), the joint distribution of the previously available coordinates does not change. Formally, for any $d' > d$, let $\pi_{d,d'} : \mathbb{R}^{d'+1} \to \mathbb{R}^{d+1}$ be the coordinate projection $\pi_{d,d'}(s_0, \dots, s_{d'}) = (s_0, \dots, s_d)$. Consistency then means that the joint distribution of $(\theta, S^{(d)})$ coincides with the distribution of $(\theta, \pi_{d,d'}(S^{(d')}))$. Equivalently, under the law $(\theta, S^{(d')})$, the marginal distribution of $(\theta, s_0, \dots, s_d)$ is identical to the distribution of $(\theta, S^{(d)})$.

Consistency implies free disposal of additional signals at the level of feasible rules. For any learning rule ℓ defined at dimension d and any $d' > d$, we can replicate the same rule at level d' via

$$\ell^+(S^{(d')}) \;:=\; \ell\big(\pi_{d,d'}(S^{(d')})\big) \;=\; \ell(S^{(d)}),$$

by ignoring the additional coordinates $s_{d+1}, \dots, s_{d'}$.

Processing Cost. Implementing a learning rule in an environment of size d induces a processing cost

$$C(\ell(S^{(d)}); d) \;\in\; [0, \infty],$$

[2] One may interpret s_0 as an "own" signal and $(s_1, \dots, s_d)$ as d additional sources. While the analysis does not rely on this interpretation, we explore this possibility in Section 6 where we recast the environment in a social learning context.

where allowing for $+\infty$ permits the inclusion of feasibility constraints such as cognitive limits or other restrictions by making some rules too costly to apply.[3]

Decision Problem. The agent chooses a learning rule ℓ to minimize the expected squared error in addition to the processing cost of implementing the chosen rule:

$$\min_{\ell(S^{(d)})\in\mathcal{L}^{(d)}} \underbrace{E\left[(\ell(S^{(d)})-\theta)^2\right]}_{\text{Inaccuracy}} + \underbrace{C(\ell(S^{(d)});d)}_{\text{Processing cost}}. \tag{P_d}$$

Given d, when an optimal learning rule exists, we denote it as $\ell_d^*(S^{(d)})$. We measure *accuracy* by the mean squared error under the optimal rule

$$\mathrm{MSE}(d) := E\left[(\ell_d^*(S^{(d)})-\theta)^2\right].$$

Note that while consistency ensures that any lower-dimensional rule can be replicated in a richer information environment with the same mean squared error, it does not preserve the total objective in (P_d). In general, $C(\ell^+;d') \neq C(\ell;d)$, so implementing the same rule may be more costly in a higher-dimensional environment.

3 Information Overload

We impose two assumptions that characterize information overload.

Our first assumption provides a "no-learning" benchmark and ensures that larger informational environments do not restrict feasible behavior.

Assumption 1 (Costless Constant Rules)**.** *If $\ell(S^{(d)}) \equiv c$ for some constant $c \in \mathbb{R}$, then $C(\ell;d) = 0$ for all d.*

Assumption 1 provides a baseline behavior: the agent can always choose a constant estimate without incurring processing costs.

Since the constant rule $\ell(S^{(d)}) \equiv \mu_\theta$ yields

$$E[(\mu_\theta-\theta)^2] = \mathrm{Var}(\theta) = \sigma_\theta^2 \quad \text{and} \quad C(\mu_\theta;d) = 0,$$

it follows that the minimized objective value in (P_d) is always weakly bounded above by σ_θ^2,

[3]Note that, unlike Rational Inattention models, where information not processed is costless, here the cost depends on the size of the information environment itself.

and the optimal MSE is uniformly bounded:

$$\mathrm{MSE}(d) \leq \sigma_\theta^2 \qquad \text{for all } d \geq 0. \tag{1}$$

This inequality ensures that, no matter the dimension of the information environment, the agent can always revert to the prior mean by following a constant rule. This reversion incurs no cost and guarantees a baseline loss that matches the variance of the prior. Moreover, since the prior is already known, mean reversion effectively implies that no learning takes place.

We next introduce the key force that drives information overload.

Standard arguments imply that if processing costs do not grow with the dimensionality of the information environment, additional signals weakly improve statistical accuracy, since irrelevant information can be freely ignored. Information overload, however, requires a force that eventually makes the use of additional information unattractive. The following provides a reduced-form condition for this to occur.

Assumption 2 (Asymptotic Cost Dominance)**.** *For any sequence of learning rules* $\{\ell_d\}_{d\geq 0}$ *with* $\ell_d(S^{(d)}) \in \mathcal{L}^{(d)}$, *if*

$$\liminf_{d\to\infty} \mathrm{Var}\big(\ell_d(S^{(d)})\big) \; > \; 0,$$

then

$$\lim_{d\to\infty} C(\ell_d; d) \; = \; +\infty.$$

Assumption 2 states that any sequence of rules that remains *informatively responsive* (i.e., $\liminf_{d\to\infty} \mathrm{Var}(\ell_d(S^{(d)})) > 0$) becomes prohibitively costly as d grows. This is not a constraint on how many signals the agent can observe. Rather, it is a constraint on the processing cost involved in producing a non-degenerate estimate in high-dimensional environments.

This is the central distinction of our framework: processing costs scale with the *dimensionality of the information environment*, rather than only with the complexity of the chosen rule. A learning rule may be *effectively low-dimensional*—relying on a small subset of cues or low-dimensional statistics—even when the environment presents a high-dimensional *menu* of signals. If costs depended only on the dimensions actively used, expanding the menu would be innocuous, as unused signals could be ignored at no additional cost.

In many environments, however, ignoring information is not costless. Experimental and process-tracing evidence shows that irrelevant information can increase decision time, while richer choice environments induce screening, multistage decision-making, and simplified weighting of cues (Chadd et al., 2021; Payne, 1976). Since cues may be unlabeled, redundant, or spurious, even simple inference procedures require screening and validation. Our framework captures this *menu burden* in reduced form by allowing costs to increase with the overall dimensionality d, even when the chosen estimator is simple.[4]

We define information overload as a global decline in accuracy beyond a finite dimensional threshold.

Definition 1 (Information Overload)**.** *Information overload occurs if there exists a finite d_1 such that for every $d_2 > d_1$,*

$$\mathrm{MSE}(d_2) \; > \; \mathrm{MSE}(d_1).$$

Our definition adopts a strong notion of overload. It requires a strict decline in accuracy, not merely diminishing marginal returns. In particular, beyond some threshold, richer environments yield strictly worse accuracy than simpler ones. The definition does not impose a specific shape on $\mathrm{MSE}(d)$: it may be flat or non-monotone at intermediate dimensions, as long as sufficiently large environments reduce accuracy. This stronger notion is appropriate in our setting, as it isolates environments in which additional information is not only unhelpful but actively harmful to performance, and thus cannot be rationalized by standard diminishing-returns arguments. This interpretation aligns with the literature emphasizing that excessive information can impair performance by exceeding processing capacity (Eppler and Mengis, 2004).[5]

To understand when this structural failure occurs, we look at the tension between statistical opportunity and processing costs when the information dimension is extremely high. Our argument focuses on the agent's performance at the limit. We show that when costs become overwhelming, the rational response to infinite dimensions is to disregard signals, which pushes the mean squared error (MSE) back to the no-learning benchmark.

The first step in establishing this behavior is to show that any learning rule that improves the MSE relative to the prior benchmark requires an increase in variance.

Lemma 1. *Fix $\varepsilon > 0$ and let ℓ be any square-integrable random variable. If*

$$E[(\ell - \theta)^2] \; \leq \; \mathrm{Var}(\theta) - \varepsilon,$$

[4]This contrasts with sparsity-based bounded rationality models (e.g., Gabaix, 2014) in which costs are tied to the effective dimension (sparsity) of the chosen representation. In such formulations, an agent can typically replicate a sparse rule in a larger environment without incurring additional cost from extra unused coordinates.

[5]In Section 5 we study a quadratic-Gaussian benchmark in which $\mathrm{MSE}(d)$ is U-shaped (equivalently, accuracy is single-peaked); see Figure 1. This special case matches the commonly observed inverted-U pattern in empirical discussions of information load and performance.

then

$$\mathrm{Var}(\ell) \;\geq\; \frac{\varepsilon^2}{4\,\mathrm{Var}(\theta)}.$$

This result links *statistical improvement* to *responsiveness*. Since the constant rule $\ell \equiv \mu_\theta$ achieves an MSE of $\mathrm{Var}(\theta)$, any nontrivial improvement requires the estimate to exhibit nontrivial variance. Thus, improving upon the prior baseline necessitates responsiveness to signals, and the lemma quantifies the minimum variance required to achieve this.

Combining Lemma 1 with Assumptions 1–2 implies that, in sufficiently complex environments, it is optimal to attenuate responsiveness and eventually revert to a prior-based constant rule.

Proposition 1. *Suppose Assumptions 1–2 hold and an optimal learning rule exists for each d. Then,*

$$\lim_{d\to\infty} \mathrm{MSE}(d) \;=\; \mathrm{Var}(\theta).$$

Proposition 1 isolates the *collapse* component of overload: regardless of the signal structure, sufficiently high dimensionality forces optimal accuracy back to the no-learning benchmark.

Recall, information overload additionally requires that the agent learns *something* at some finite dimension level.

Proposition 2. *Suppose Assumptions 1–2 hold and an optimal learning rule exists for each d. Information overload occurs if and only if both of the following conditions hold:*

(i) *(*Initial informativeness*) There exists a finite d_0 such that* $\mathrm{MSE}(d_0) < \mathrm{Var}(\theta)$*.*

(ii) *(*Asymptotic collapse*)* $\lim_{d\to\infty} \mathrm{MSE}(d) = \mathrm{Var}(\theta)$*.*

Proposition 2 shows that information overload operates through a stark non-monotonicity. At moderate informational dimensions, agents lie in a *learning region*, where responsiveness to signals improves accuracy relative to the prior benchmark. Beyond a threshold, however, rising complexity pushes the environment into an *overload region* where optimal responsiveness collapses and accuracy reverts to the benchmark despite the availability of additional information.

Several distinct mechanisms can generate this kind of asymptotic collapse.

First, in many high-dimensional inference problems, achieving nontrivial accuracy requires estimators whose effective dimension grows with d (Tibshirani, 1996; Wainwright, 2019). Even when the target parameter is low-dimensional, valid estimation may require learning high-dimensional nuisance components whose complexity increases with the environment

(Laan and Rose, 2011; Newey and McFadden, 1994). If agents face constraints on implementable complexity, the set of informative rules may shrink with d, leaving only low-responsiveness rules.

Second, collapse can arise from simplified or misspecified representations. For example, correlation neglect may emerge when agents approximate the true distribution with parsimonious models that impose independence restrictions (Spiegler, 2016). As a result, agents become less responsive to additional information that their representation cannot fully exploit.

Third, collapse can occur even under correct specification if responsiveness is intrinsically costly. This is the mechanism emphasized in rational inattention models, where capacity costs induce agents to coarsen their information and attenuate responsiveness (Sims, 2003). In the remainder of the paper, we focus on this channel and interpret processing costs as *cognitive costs* that increase with responsiveness.

Lastly, other than proving necessary and sufficient conditions for overload to arise, the result also suggests a broader interpretation of information overload. It shows that as the informational environment becomes sufficiently high-dimensional, rational learning need not become richer or more sophisticated. Instead, the optimal response may become behaviorally simpler. Agents attenuate responsiveness to incoming signals and rely increasingly on prior-based or less responsive rules. In this sense, behavior that may appear heuristic-like in information-rich environments need not reflect a departure from rationality.

4 Learning Structure

Section 3 established that, under asymptotic cost dominance, accuracy must eventually collapse as the informational dimension grows. That argument relies only on the existence of optimal learning rules and not their particular form. To gain further insight, we proceed to characterize the structure of optimal learning under two standard restrictions. This will, in turn, allow us to study how accuracy varies with informational dimension in Section 5.

Throughout, fix an information level $d \geq 0$ and write $S^{(d)} \in \mathbb{R}^{d+1}$ for the signal vector.

4.1 A Convex-Analytic Formulation

To characterize the structure of optimal learning, we begin by providing some additional structure on (P_d).

Recall that $\mathcal{L}^{(d)}$ denotes the set of square-integrable estimates that can be generated from

$S^{(d)}$. Endow $\mathcal{L}^{(d)}$ with the inner product

$$\langle f, g \rangle := E[f(S^{(d)})g(S^{(d)})],$$

and induced norm $\|f\|_2 := \sqrt{\langle f, f \rangle}$. This makes $(\mathcal{L}^{(d)}, \langle \cdot, \cdot \rangle)$ a Hilbert space.

Defining the functional

$$J_d(\ell) := E\big[(\ell(S^{(d)}) - \theta)^2\big] + C(\ell; d), \qquad \ell \in \mathcal{L}^{(d)},$$

then allows us to recast (P_d) as a functional minimization problem on a Hilbert space.

This formulation is useful for two reasons. First, it clarifies that what the agent is choosing is a measurable mapping from $\mathbb{R}^{d+1}$ to $\mathbb{R}$. Second, it allows us to identify a standard set of conditions under which the minimization problem is well-posed and can be studied using convex analysis.

Assumption 3. *For each $d \geq 0$, the cost functional $C(\cdot; d) : \mathcal{L}^{(d)} \to [0, \infty]$ is:*

(i) Proper: *there exists some $\ell \in \mathcal{L}^{(d)}$ such that $C(\ell; d) < \infty$;*

(ii) Convex: *for all $\ell_1, \ell_2 \in \mathcal{L}^{(d)}$ and $\alpha \in [0, 1]$,*

$$C(\alpha\ell_1 + (1 - \alpha)\ell_2; d) \leq \alpha C(\ell_1; d) + (1 - \alpha)C(\ell_2; d);$$

(iii) Lower semicontinuous: *if $\ell_n \to \ell$ in $\|\cdot\|_2$, then*

$$C(\ell; d) \leq \liminf_{n\to\infty} C(\ell_n; d).$$

Assumption 3 is standard in nonparametric choice problems. Properness rules out degenerate cases in which all rules are infeasible. Convexity captures diminishing returns in cognitive effort, as mixtures of rules are no more costly than the average of their costs. Lower semi-continuity ensures that small perturbations of a rule do not induce discontinuous drops in cost.

Proposition 3. *Under Assumption 3, for each $d \geq 0$, the problem (P_d) admits a unique minimizer $\ell_d^* \in \mathcal{L}^{(d)}$ (unique up to almost-sure equality).*

Proposition 3 guarantees that the agent's problem is well-defined at each dimension d, allowing us to study the paths $d \mapsto \ell_d^*$ and $d \mapsto \mathrm{MSE}(d)$.

4.2 Affine Optimality

Let $\mathcal{A}^{(d)}$ denote the set of affine functions of d-dimensional signal vectors:

$$\mathcal{A}^{(d)} := \left\{ a \in \mathcal{L}^{(d)} \mid a(S^{(d)}) = \omega^\top S^{(d)} + c \text{ for some } (\omega, c) \in \mathbb{R}^{d+1} \times \mathbb{R} \right\}.$$

This is a finite-dimensional linear subspace of $\mathcal{L}^{(d)}$ (hence closed).

We now introduce the two restrictions that deliver affine rules as the optimal learning structure.

Assumption 4. *For each $d \geq 0$, the conditional expectation of the state given the signal vector is affine:*

$$E[\theta \mid S^{(d)}] \in \mathcal{A}^{(d)}.$$

Assumption 4 is satisfied in the workhorse benchmark where $(\theta, S^{(d)})$ is jointly Gaussian, and more generally for elliptical families. It states that the statistically optimal predictor (the L^2 projection of θ onto the sigma-algebra generated by $S^{(d)}$) is affine in the signals. This rules out environments in which extracting information necessarily requires higher-order nonlinear features of the data. Importantly, this assumption does *not* impose affine learning; it only restricts the form of the best predictor.[6]

Assumption 5. *For each $d \geq 0$, the cost functional is (weakly) increasing in the variance of the induced estimate: for any $f, g \in \mathcal{L}^{(d)}$,*

$$\mathrm{Var}\big(f(S^{(d)})\big) \geq \mathrm{Var}\big(g(S^{(d)})\big) \quad \Longrightarrow \quad C(f; d) \geq C(g; d).$$

Assumption 5 formalizes the idea that, holding the dimension d fixed, learning rules that generate greater variability in the estimate are (weakly) more cognitively costly. The assumption does not impose differentiability or any parametric structure. It allows for arbitrary nondecreasing mappings from induced variance into cost, which may vary with d. Costs thus depend on the rule only through its *responsiveness*, as measured by variance, together with the dimensionality of the information environment.[7]

Proposition 4. *Suppose Assumptions 3–5 hold. Then, the unique minimizer ℓ_d^* of (P_d) belongs to $\mathcal{A}^{(d)}$. Equivalently, there exist $(\omega_d^*, c_d^*) \in \mathbb{R}^{d+1} \times \mathbb{R}$ such that*

$$\ell_d^*(S^{(d)}) = (\omega_d^*)^\top S^{(d)} + c_d^*.$$

[6] In the absence of cognitive costs, the optimal estimator is $\mathbb{E}[\theta \mid S^{(d)}]$.

[7] In Section 5, we present a quadratic specification that pins down this ranking sharply.

Proposition 4 provides a rational justification for the use of affine learning rules, which are often imposed as behavioral primitives (e.g., DeGroot (1974)). The result is best understood as a dominance argument. Any learning rule can be decomposed into an affine component $a(S^{(d)})$ and a residual $r(S^{(d)})$ which is orthogonal to $\mathcal{A}^{(d)}$. The residual does not improve predictive accuracy about θ beyond what is already captured by the affine conditional expectation, but increases dispersion in the realized estimate. Under variance-monotone costs, this makes r weakly harmful for both accuracy and cost, so it is optimally set to zero.

A key implication of Proposition 4 is that it reduces the infinite-dimensional problem of choosing a measurable rule $\ell(\cdot)$ to a finite-dimensional problem over coefficients $(\omega, c) \in \mathbb{R}^{d+1} \times \mathbb{R}$.

Corollary 1. *Suppose Assumptions 3–5 hold. Define the induced cost on coefficients*

$$\widetilde{C}_d(\omega, c) := C\big(\omega^\top S^{(d)} + c;\, d\big).$$

Then (P_d) *is equivalent to the convex program*

$$\min_{(\omega,c)\in\mathbb{R}^{d+1}\times\mathbb{R}} \; E\big[(\omega^\top S^{(d)} + c - \theta)^2\big] + \widetilde{C}_d(\omega, c),$$

and any optimizer (ω_d^*, c_d^*) *yields the unique optimal learning rule* $\ell_d^*(S^{(d)}) = (\omega_d^*)^\top S^{(d)} + c_d^*$.

Corollary 1 shows that the problem is fully summarized by the coefficients (ω, c). This highlights the distinction between the cost of *using* information and the cost of *processing* the information environment. Standard attention models (e.g., Gabaix (2014)) emphasize effective dimension by penalizing the magnitude or sparsity of the weights ω, capturing how strongly agents respond to signals. Our framework instead captures a class of responsiveness costs that depend on the induced dispersion $\mathrm{Var}(\omega^\top S^{(d)})$, allowing cognitive effort to be represented as a nondecreasing function of variance.[8]

At the same time, we allow for a distinct menu burden. As the information environment expands, the cognitive load of screening and validating potential signals can increase, even if the agent ultimately assigns a nonzero weight to only a few coordinates. Formally, this is why the induced cost $\widetilde{C}_d(\omega, c)$ is allowed to depend explicitly on d in addition to (ω, c). The overload result, therefore, does not arise from restricting the agent to complex rules, but from the fact that identifying a small set of useful signals within a large and noisy environment is itself costly.

[8] Sparsity penalties such as $\sum_i |\omega_i|$ need not be monotone in $\mathrm{Var}(\omega^\top S^{(d)})$ and thus are not implied by Assumption 5; we view them as an analogous finite-dimensional formulation that shares the "penalize responsiveness" intuition.

5 Quadratic Cognitive Costs

Sections 3–4 established two general results. First, under asymptotic cost dominance, inaccuracy eventually returns to the no-learning benchmark as the informational dimension grows. Second, under affine conditional expectations and variance-monotone costs, the optimal learning rule is affine.

This section combines these insights in a tractable benchmark that delivers closed-form solutions and transparent comparative statics. This benchmark has two features: (i) a quadratic, variance-based processing cost, and (ii) a linear Gaussian signal structure. The key mechanism is that, as the information environment expands, the agent attenuates responsiveness to manage cognitive costs, generating a U-shaped pattern of inaccuracy as a function of informational load.

Throughout, fix an information level $d \geq 0$ and write $S^{(d)} = (s_0, \ldots, s_d)^\top$.

5.1 Variance-Based Cost and Gaussian Priors

We adopt a quadratic cost that penalizes the dispersion of the estimate across realizations of the signal vector:

$$C_r(\ell(S^{(d)}); d) \;:=\; \lambda(d+1)\,\mathrm{Var}\big(\ell(S^{(d)})\big), \tag{2}$$

where $\lambda \geq 0$ controls the cognitive cost of being responsive. If $\lambda = 0$, the agent does not face any cognitive costs. In this situation, as we demonstrate below, the agent can process the signals in a fully Bayesian way.

The quantity $\mathrm{Var}(\ell(S^{(d)}))$ measures the responsiveness of the estimate to the data. Constant (prior-based) rules have zero variance, while more responsive rules generate more dispersed estimates. The $(d+1)$ factor captures scaling costs: processing higher-dimensional inputs requires additional cognitive effort.

Because $C_r(\cdot; d)$ depends on ℓ only through $\mathrm{Var}(\ell(S^{(d)}))$, it is variance-monotone and convex on L^2. It follows that this cost function satisfies Assumptions 3 and 5.[9] Therefore, the optimal rule under $C_r(\cdot; d)$ is affine. The remainder of the section thus focuses on characterizing the optimal affine coefficients and their properties.

With regard to information, we assume that the state and signals satisfy:

[9] We verify that C_r also satisfies the structural assumptions of Section 3 and 4. Assumption 1: If $\ell \equiv c$, then $\mathrm{Var}(\ell) = 0$, so $C_r(\ell; d) = 0$. Assumption 2: if $\mathrm{Var}(\ell) > 0$, then $C_r \to \infty$ as $d \to \infty$. Assumption 3: The functional $\ell \mapsto \mathrm{Var}(\ell) = \|\ell - E[\ell]\|_2^2$ is convex and continuous (hence lower semicontinuous) with respect to the L^2 norm. Assumption 5: Since $\lambda(d+1) \geq 0$, $C_r(\ell; d)$ is nondecreasing in $\mathrm{Var}(\ell)$.

Assumption 6. *The state and signals satisfy:*

$$\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2),$$
$$s_k = \theta + \varepsilon_k, \qquad k = 0, 1, \ldots, d,$$
$$(\varepsilon_k)_{k=0}^d \text{ i.i.d. } \mathcal{N}(0, \sigma_\varepsilon^2), \quad \text{independent of } \theta.$$

Note that Assumption 6 is imposed only to obtain closed-form expressions. The mechanism identified in Section 4 does not rely on normality; rather, normality allows us write the optimal rule and the corresponding MSE in a transparent way.[10]

5.2 Closed-Form Optimal Rule

Let $\mathbf{1}_{d+1} \in \mathbb{R}^{d+1}$ denote the vector of ones.

Proposition 5. *Under Assumption 6 and the quadratic cost, the unique optimizer of* (P_d) *is affine,*

$$\ell_d^*(S^{(d)}) = (\omega_d^*)^\top S^{(d)} + c_d^*,$$

with symmetric weights $\omega_d^* = \alpha_d \, \mathbf{1}_{d+1}$*, where*

$$\alpha_d = \underbrace{\frac{1}{1 + \lambda(d+1)}}_{\text{cognitive deflator}} \cdot \underbrace{\frac{\sigma_\theta^2}{\sigma_\varepsilon^2 + \sigma_\theta^2(d+1)}}_{\text{Bayesian per-signal weight}}, \tag{3}$$

$$c_d^* = \Big(1 - (\omega_d^*)^\top \mathbf{1}_{d+1}\Big)\mu_\theta. \tag{4}$$

Equivalently, let $\bar{s}^{(d)} := \frac{1}{d+1}\sum_{k=0}^d s_k$*, then*

$$\ell_d^*(S^{(d)}) = \mu_\theta + \kappa_d\big(\bar{s}^{(d)} - \mu_\theta\big), \qquad \kappa_d := (d+1)\alpha_d = \frac{1}{1+\lambda(d+1)} \cdot \frac{\sigma_\theta^2(d+1)}{\sigma_\varepsilon^2 + \sigma_\theta^2(d+1)}. \tag{5}$$

Equation (3) separates two forces. The second factor is the standard Bayesian precision term. If $\lambda = 0$, each signal receives the usual Bayesian weight. The first factor, $(1 + \lambda(d + 1))^{-1}$, is a *cognitive deflator* that uniformly shrinks all weights as either cognitive costs λ or dimensionality d increases. Relative to the frictionless benchmark, the agent attenuates responsiveness to all signals.

We proceed to discuss three implications of Proposition 5.

[10] Assumption 6 implies Assumption 4.

Distribution and Variance. Since ℓ_d^* is an affine function of jointly Gaussian variables, it is also Gaussian. Moreover, the estimator is prior-unbiased ($E[\ell_d^* - \theta] = 0$) and its variance admits a simple form.

Corollary 2. *Suppose Assumption 6 and the quadratic cost specification hold. Then*

$$\ell_d^*(S^{(d)}) \sim \mathcal{N}\Big(\mu_\theta,\ \mathrm{Var}(\ell_d^*(S^{(d)}))\Big),$$

with

$$\mathrm{Var}(\ell_d^*(S^{(d)})) = \kappa_d^2\Big(\sigma_\theta^2 + \frac{\sigma_\varepsilon^2}{d+1}\Big).$$

Relative to the case $\lambda = 0$, the variance is uniformly smaller since κ_d decreases with λ. This reflects shrinkage toward the prior induced by cognitive costs.

Closed-Form MSE. Accuracy is measured by the optimal mean squared error $\mathrm{MSE}(d) = E[(\ell_d^*(S^{(d)}) - \theta)^2]$, which decomposes into a biased-prior component and a noise component.

Corollary 3. *Suppose Assumption 6 and the quadratic cost specification hold. Then*

$$\mathrm{MSE}(d) = \big(1 - \kappa_d\big)^2\sigma_\theta^2 + \kappa_d^2\frac{\sigma_\varepsilon^2}{d+1}. \tag{6}$$

The first term captures the bias from shrinkage toward the prior (since $\kappa_d < 1$), while the second term reflects residual sampling noise from averaging signals.

Asymptotic Collapse. The cognitive deflator forces $\kappa_d \to 0$ as $d \to \infty$ when $\lambda > 0$, producing an explicit instance of the general collapse result from Section 3. Furthermore, as $\lambda \to \infty$, $\kappa_d \to 0$, implying that a high cognitive coefficient induces the same collapse effect.

Corollary 4. *Suppose Assumption 6 and the quadratic cost specification hold. Then,*

(i) If $\lambda > 0$, then $\kappa_d \to 0$ as $d \to \infty$, and hence

$$\ell_d^*(S^{(d)}) \to \mu_\theta \text{ in } L^2, \qquad \mathrm{MSE}(d) \to \sigma_\theta^2.$$

(ii) If $\lambda = 0$, then $\kappa_d \to 1$ as $d \to \infty$, and $\mathrm{MSE}(d) \to 0$.

Part (i) recovers the no-learning benchmark: in sufficiently high-dimensional environments, the agent optimally attenuates responsiveness and reverts to the prior mean. Part (ii) describes the frictionless benchmark in which additional information always improves accuracy.

5.3 Optimal Dimension

To connect our closed-form solution to the overload narrative, it is useful to separate *accuracy* (or statistical performance) as measured by MSE(d), from the *total loss* which quantifies the minimized value of (P_d) and which includes cognitive costs. At information level d, denote that latter by

$$L(d) := \min_{\ell \in \mathcal{L}^{(d)}} E[(\ell(S^{(d)}) - \theta)^2] + C_r(\ell; d) = \mathrm{MSE}(d) + \lambda(d+1)\,\mathrm{Var}(\ell_d^*(S^{(d)})).$$

Equations (5) and (6) imply that changes in d affect $L(d)$ through two channels: (i) an *information effect*, whereby additional signals reduce noise in the average signal and improve accuracy, and (ii) a *cognitive load effect*, whereby higher dimensionality raises the marginal cost of responsiveness, reducing κ_d and increasing bias.

The following result provides a full characterization of the behavior of $L(d)$ as we vary the dimension of the information environment.

Proposition 6. *Under Assumption 6 and the quadratic cost, the total loss $L(d)$ changes with d as follows:*[11]

(i) If $\lambda > \frac{\sigma_\varepsilon^2}{2\sigma_\theta^2}$, then $L(d)$ is strictly increasing in d.

(ii) If $\lambda \le \frac{\sigma_\varepsilon^2}{2\sigma_\theta^2}$, then there exists an integer $d^* \ge 0$ such that $L(d)$ is strictly decreasing for $d < d^*$ and strictly increasing for $d > d^*$.

The cutoff between corner and interior solutions is governed by the relative magnitude of cognitive costs and statistical gains. In particular, Proposition 6 implies that the optimal dimension collapses to $d^* = 0$ if and only if the cognition parameter λ is sufficiently large relative to the signal-to-prior ratio $\sigma_\varepsilon^2/\sigma_\theta^2$. Otherwise, the optimal choice is interior. In that case, even if the signals are noisier, the value of aggregating additional independent signals increases because averaging reduces idiosyncratic noise. At the same time, as d increases, the weight placed on each signal declines, so the agent becomes less responsive to any particular source even as the optimal information set expands.

While Proposition 6 characterizes optimal behavior in terms of total loss $L(d)$, information overload is defined in terms of statistical performance, i.e., MSE(d). However, the same force that eventually makes additional signals unattractive in terms of total loss (rising cognitive costs that induce stronger shrinkage) also drives the deterioration of accuracy at high dimensions. Specifically, Corollary 4 implies that $\kappa_d \to 0$ as $d \to \infty$, so MSE(d) $\to \sigma_\theta^2$.

[11] If $\lambda = 0$, then $L(d)$ is strictly decreasing in d and there is no finite minimizer (equivalently, $d^* = \infty$). Henceforth assume $\lambda > 0$.

Thus, if learning improves accuracy at intermediate dimensions, MSE(d) must eventually increase and return to the no-learning benchmark, generating the pattern illustrated in Figure 1.

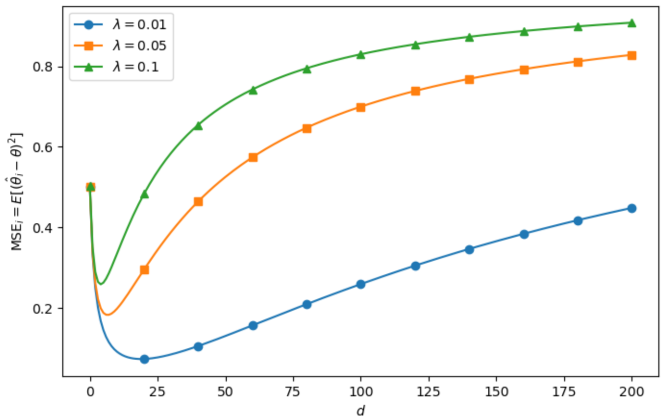

Figure 1: A U-shaped inaccuracy curve (MSE) under quadratic costs. Illustration assumes $\sigma_\theta^2 = \sigma_\varepsilon^2$.

This U-shaped pattern is a clear instance of information overload as defined in Definition 1. It corresponds to the familiar inverted-U relationship between information and accuracy documented in the overload literature (Eppler and Mengis, 2004; Roetzel, 2019). While that literature focuses on accuracy, our model is expressed in terms of inaccuracy (MSE), making the two representations equivalent.

This pattern is also consistent with the findings of Dasaratha and He (2021), who show that agents achieve higher accuracy in sparse networks than in dense ones. Interpreting d as network degree, sparse networks reduce cognitive burden, while dense networks increase it. We explore this interpretation further in Section 6.

5.4 Heterogeneous Signal Quality

We now extend the model to allow for heterogeneous signal quality and show that the overload mechanism does not rely on symmetric signals: even as cognitive costs reduce overall responsiveness, the agent continues to assign relatively higher weight to more precise signals.

Assumption 7. *The state satisfies $\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$ and the signals satisfy $s_k = \theta + \varepsilon_k$ with independent Gaussian noise*

$$\varepsilon_k \sim \mathcal{N}(0, \sigma_{\varepsilon,k}^2), \qquad k = 0, 1, \ldots, d,$$

independent of θ.

Let $P_d := \sum_{k=0}^{d} \sigma_{\varepsilon,k}^{-2}$ denote total precision.

Proposition 7. *Under Assumption 7 and the quadratic cost, the unique optimal rule is affine*

$$\ell_d^*(S^{(d)}) = (\omega_d^*)^\top S^{(d)} + c_d^*, \qquad c_d^* = \big(1 - (\omega_d^*)^\top \mathbf{1}_{d+1}\big)\mu_\theta,$$

with weight components

$$\omega_{d,k}^* = \frac{1}{1+\lambda(d+1)} \cdot \frac{\sigma_\theta^2}{1+\sigma_\theta^2 P_d} \cdot \frac{1}{\sigma_{\varepsilon,k}^2}, \qquad k = 0, 1, \ldots, d.$$

Moreover, for any two sources j, k,

$$\sigma_{\varepsilon,k}^2 < \sigma_{\varepsilon,j}^2 \iff \omega_{d,k}^* > \omega_{d,j}^*.$$

These optimal weights separate two roles. The common factor $(1 + \lambda(d + 1))^{-1}$ controls *overall responsiveness*, shrinking all weights as cognitive costs or dimensionality increase. In contrast, heterogeneity in $\sigma_{\varepsilon,k}^2$ governs *relative allocation*, with more precise signals receiving higher weight.

Broadly, the result reflects the intuition of trust and selective attention in social learning. It shows that a rational agent balancing accuracy and cognitive cost will endogenously discount noisy sources and prioritize reliable ones. This is because the weight ω_k is inversely proportional to the noise of signal k, $\sigma_{\varepsilon,k}^2$.

It is useful, therefore, to distinguish the roles of two parameters λ and $\sigma_{\varepsilon,k}^2$. Parameter λ governs overall responsiveness: higher cognitive costs reduce attention to all signals. In contrast, heterogeneity in $\sigma_{\varepsilon,k}^2$ determines how attention is allocated across signals. Thus, even under cognitive constraints, agents remain selective, placing greater weight on more reliable sources. Total responsiveness remains endogenous and depends jointly on cognitive costs and signal quality.

6 Applications

We apply our framework to two settings. First, we show that information overload can give rise to correlation neglect-like behavior. Second, we provide a network interpretation of the information dimension d by mapping it to network degree and derive implications for social learning.

6.1 Correlation Neglect

A robust experimental finding is that decision makers often treat redundant signals as if they were independent, failing to discount information that is correlated across sources (e.g., Dasaratha and He, 2021; Enke and Zimmermann, 2019). This phenomenon is known as *correlation neglect* and is commonly interpreted as a behavioral deviation from statistically correct aggregation.

Our framework offers a complementary interpretation. In high-dimensional environments, cognitive costs induce strong shrinkage toward the prior, reducing the benefits of modeling correlation precisely. While we do not derive correlation neglect as an endogenous choice, we show that as dimensionality increases, learning rules that neglect correlation and those that internalize it become behaviorally equivalent. Thus, behavior akin to correlation neglect can arise as a consequence of optimal attenuation of responsiveness under high informational load.

Set Up

We model redundancy by allowing the idiosyncratic noises to be positively correlated across signals. Let

$$\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2), \qquad s_k = \theta + \varepsilon_k, \quad k = 0, 1, \ldots, d,$$

where the noise vector $\varepsilon := (\varepsilon_0, \ldots, \varepsilon_d)^\top$ is independent of θ and distributed as

$$\varepsilon \sim \mathcal{N}\big(0,\ \sigma_\varepsilon^2 \Sigma_\rho\big)\,.$$

We now assume that the correlation matrix Σ_ρ is equicorrelated:

$$\Sigma_\rho := (1-\rho)I + \rho\,\mathbf{1}\mathbf{1}^\top, \qquad \rho \in (0,1).$$

Consequently, the signals $s_k = \theta + \varepsilon_k$ share a common component and correlated noise; larger ρ makes additional signals more redundant (less informative at the margin) about θ.[12]

Let $S^{(d)} = (s_0, \ldots, s_d)^\top$. Under this information structure, the covariance matrix of $S^{(d)}$ is

$$\mathrm{Var}(S^{(d)}) = \sigma_\theta^2\,\mathbf{1}\mathbf{1}^\top + \sigma_\varepsilon^2\big[(1-\rho)I + \rho\,\mathbf{1}\mathbf{1}^\top\big].$$

[12] In particular, $\mathrm{Var}(s_k) = \sigma_\theta^2 + \sigma_\varepsilon^2$ and $\mathrm{Corr}(s_i, s_j) = (\sigma_\theta^2 + \rho\sigma_\varepsilon^2)/(\sigma_\theta^2 + \sigma_\varepsilon^2)$.

We maintain the quadratic processing cost benchmark used in Section 5:

$$C_r(\ell; d) = \lambda(d+1)\,\mathrm{Var}(\ell(S^{(d)})), \qquad \lambda \geq 0.$$

By the affine optimality result in Section 4, the optimal rule is affine. Moreover, under equicorrelation, symmetry implies that all signals receive the same weight.

Remark 1. *Maintain the Gaussian environment above and the quadratic processing cost. The optimal learning rule takes the form*[13]

$$\ell^*(S^{(d)}) = (\omega^*)^\top S^{(d)} + c^*, \qquad \omega^* = \alpha^* \mathbf{1}, \qquad c^* = \big(1 - (\omega^*)^\top \mathbf{1}\big)\mu_\theta,$$

where

$$\alpha^* = \frac{\sigma_\theta^2}{(1+\lambda(d+1))\Big(\sigma_\varepsilon^2(1-\rho) + (d+1)(\sigma_\theta^2 + \sigma_\varepsilon^2\rho)\Big)}.$$

The expression highlights the redundancy channel: for fixed d, a higher ρ increases effective noise in the signal vector, leading the agent to reduce weights on each signal.

Correlation Neglect as a Misspecified Rule

To model correlation neglect, consider an agent who faces the same cognitive cost but misperceives the covariance structure by treating signals as independent (i.e., sets $\rho = 0$). The agent, therefore, solves the same quadratic-cost problem under a misspecified covariance matrix. The resulting rule is again affine and symmetric.

Definition 2. *The correlation-neglect rule is the affine learning rule obtained by solving the quadratic-cost problem under the misspecified covariance structure $\rho = 0$:*

$$\ell^N(S^{(d)}) = (\omega^N)^\top S^{(d)} + c^N, \qquad \omega^N = \alpha^N \mathbf{1}, \qquad c^N = \big(1 - (\omega^N)^\top \mathbf{1}\big)\mu_\theta,$$

with

$$\alpha^N = \frac{\sigma_\theta^2}{(1+\lambda(d+1))\Big(\sigma_\varepsilon^2 + (d+1)\sigma_\theta^2\Big)}.$$

The gap between ω^* and ω^N measures the distortion induced by neglecting correlation. We now examine how this distortion behaves as the information environment expands.

[13] We can characterize this learning rule by using the same proof logic in Proposition 5.

Disappearing Gap Under Overload

The key mechanism is the cognitive deflator $(1+\lambda(d+1))^{-1}$. As dimensionality increases, responsiveness is increasingly attenuated, pushing both rules toward the prior. As a result, differences in covariance modeling have diminishing effects on behavior.

Proposition 8. *Let $\rho \in (0,1)$ and $\lambda > 0$. The discrepancy between the correlation-aware and correlation-neglect weights satisfies*[14]

$$\|\omega^* - \omega^N\|_2 = O\left(\frac{1}{\lambda(d+1)^{3/2}}\right).$$

Proposition 8 shows that the distance between the correlation-aware and correlation-neglect rules shrinks rapidly as dimensionality increases. The two rules are, therefore, behaviorally equivalent in the limit. Two forces drive this convergence: (i) as d increases, the marginal value of modeling correlation declines relative to the overall shrinkage induced by cognitive costs, and (ii) when λ is large, responsiveness is attenuated regardless of the covariance structure, pushing both rules toward the same prior-based estimate.

This yields a concrete implication. In low-dimensional environments or when cognitive costs are small, accounting for correlation matters quantitatively. By contrast, in high-dimensional environments with strong shrinkage, behavior under correlation neglect closely approximates fully rational behavior. Thus, our framework rationalizes when correlation neglect is less consequential and harder to detect, namely, in environments with many information sources or high cognitive burden. We do not claim that agents misunderstand correlation; rather, once optimal behavior is already close to the prior, the payoff-relevant consequences of modeling correlation correctly become small.

6.2 Social Learning

We now provide a network interpretation by mapping the information dimension d to an agent's degree in a social network. Agent i observes her own signal and those of her d_i neighbors, so comparative statics in d translate into comparative statics in network connectivity. This yields implications for optimal degrees, degree heterogeneity, endogenous influence weights, and the DeGroot benchmark.

[14] We assume $\lambda > 0$.

Set Up

Let $g = (N, E)$ be an undirected graph with node set $N = \{1, \ldots, n\}$. For each agent i, let

$$N_i(g) := \{j \in N : \{i, j\} \in E\} \quad \text{and} \quad d_i := |N_i(g)|$$

respectively denote the agent's neighborhood and degree. Agent i observes her own signal and those of her neighbors. Her observed signal vector is[15]

$$S_i(g) := \big(s_j\big)_{j \in N_i(g) \cup \{i\}} \in \mathbb{R}^{d_i+1}.$$

As before, a learning rule for agent i is any measurable map $\ell_i : \mathbb{R}^{d_i+1} \to \mathbb{R}$ with finite second moment under the joint distribution of $(\theta, S_i(g))$.

Given degree d_i and information $S_i(g)$, agent i solves

$$\min_{\ell_i} \; E\big[(\ell_i(S_i(g)) - \theta)^2\big] \; + \; C_r(\ell_i; d_i)\,,$$

where $C_r(\ell_i; d_i)$ is the same cost functional as in Section 5:

$$C_r(\ell_i; d_i) = \lambda_i(d_i + 1) \; \mathrm{Var}[\ell_i(S_i(g))]\,,$$

where we now allow for cognitive-cost coefficients $\lambda_i \geq 0$ to vary across agents.

This gives the following correspondence: the dimension of the information environment d faced by an agent i can be interpreted as the agent's network degree d_i once we specify that agents observe neighbors' signals.

In the homoskedastic benchmark, symmetry implies equal weights across observed signals. Letting $m_i := d_i + 1$ denote the number of observed signals, our earlier results imply that

$$\ell_i^*(S_i) = \omega_i^* \sum_{j \in N_i(g) \cup \{i\}} s_j + c_i^*,$$

where

$$\omega_i^* = \frac{1}{1 + \lambda_i m_i} \cdot \frac{\sigma_\theta^2}{\sigma_\varepsilon^2 + \sigma_\theta^2 m_i}, \qquad c_i^* = \Big(1 - \omega_i^* m_i\Big)\mu_\theta.$$

The first factor is the cognitive deflator; the second is the usual Bayesian precision weight.

[15] Thus, the dimension of i's information environment is $d_i + 1$.

Optimal Degrees and Degree Distribution

Our analysis in Section 5.3 implies that each agent has a preferred information dimension—and hence, in this interpretation, an optimal degree—that balances statistical gains from additional signals against cognitive costs.

Suppose that each agent can choose how many different sources to observe, for example, by selecting the number of social media accounts to follow or professional contacts to keep. Formally, agent i chooses an integer degree d_i to minimize the total loss resulting from the best learning rule under a signal vector of dimension $d_i + 1$.

Proposition 6 implies two regimes. If cognitive costs are sufficiently large relative to signal quality, total loss increases with d_i and the optimal degree is $d_i^* = 0$. Otherwise, total loss is U-shaped, yielding an interior optimal degree.

Focusing on the nontrivial case, suppose that optimal degrees are interior for all agents. This is guaranteed when

$$\lambda_i < \frac{\sigma_\varepsilon^2}{2\sigma_\theta^2}, \quad \forall i \in N.$$

Treating $m_i = d_i + 1$ as continuous variable, we get the unique minimizer[16]

$$m_i^* = \sqrt{\frac{\sigma_\varepsilon^2}{\lambda_i \sigma_\theta^2}}, \qquad \text{so} \qquad d_i^* \approx \sqrt{\frac{\sigma_\varepsilon^2}{\lambda_i \sigma_\theta^2}} - 1,$$

with the exact discrete optimum obtained by checking the nearest integers around this value.

The expression shows that d_i^* scales approximately as $\lambda_i^{-1/2}$. Agents with low cognitive costs choose higher degrees, while those with high cognitive costs remain in sparse environments to avoid overload. This relationship is illustrated in the figure below, which shows that the optimal degree is a decreasing function of the cognitive cost coefficient when fixing other parameters (σ_ε^2 and σ_θ^2).

[16]The optimal degree is calculated in the proof of Proposition 6.

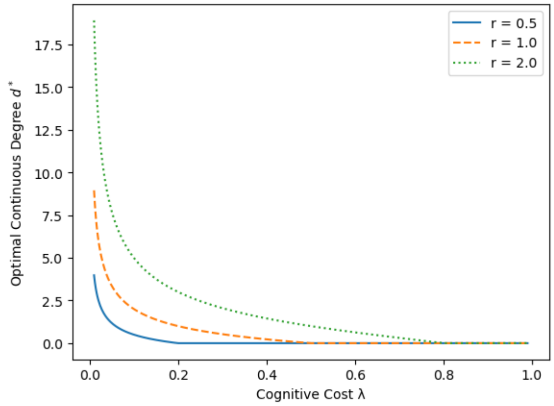

Figure 2: Optimal degree decreases in the cognitive cost λ where $r = \frac{\sigma_\varepsilon^2}{\sigma_\theta^2}$.

The figure shows that once we interpret d_i as the number of social links while assuming agents can choose degrees close to their preferred levels, then the distribution of cognitive capacity $\{\lambda_i\}_{i\in N}$ induces a degree distribution. If a high proportion of the population is subject to high cognitive costs while the rest face low costs, the mapping $\lambda_i \mapsto d_i^*$ leads to skewed degree distributions. In such a case, we would observe many individuals with low degrees and a small number of hubs with high degrees. This provides a cognition-based explanation for heavy-tailed degree distributions observed in social networks (Jackson, 2008).

Consequences for Informational Interventions

Our results also give rise to implications regarding informational intervention on an exogenous network.[17] Standard diffusion models target highly connected agents because of their centrality (Banerjee et al., 2013). In our framework, however, a higher degree implies a larger information environment and thus stronger cognitive shrinkage. As a result, holding signal quality fixed, high-degree agents place lower marginal weight on any additional signal, making them less responsive to new information.

This is a statement about responsiveness rather than full diffusion dynamics. In models where propagation depends on initial responses, our mechanism implies that targeting highly connected agents may be less effective when cognitive costs are significant. More generally, degree and influence need not align once responsiveness is endogenously attenuated.

Weighted Networks

We can further extend this social learning perspective by invoking the heterogeneity result of Proposition 7. This allows us to take into account link weights. While the underlying

[17]Some agents may then operate above their preferred degree.

contact network g stays unweighted and undirected (defining which signals are observable to agents), the optimal learning rule gives continuous, heterogeneous weights to neighbors based on their signal reliability.

Suppose signals differ in quality across neighbors:

$$s_j = \theta + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \sigma^2_{\varepsilon,j}), \qquad \varepsilon_j \perp \varepsilon_k \ \ (j \neq k),$$

and maintain the same quadratic cost. Then, Proposition 7 implies that the optimal affine rule assigns a distinct weight to each observed source

$$\ell_i^*(S_i) = \sum_{j \in N_i(g) \cup \{i\}} \omega_{ij}^* s_j + c_i^*, \qquad \omega_{ij}^* \propto \frac{1}{\sigma^2_{\varepsilon,j}},$$

where ω_{ij} is the j-element of a vector ω_i with a common multiplicative shrinkage term driven by (λ_i, d_i).

This expression shows that while the unweighted network g determines which signals are observable, the endogenous weights $\{\omega_{ij}^*\}$ can be overlaid on g to give a weighted influence network. The entry ω_{ij}^* measures the extent to which neighbor j influences agent i's belief. This construction can thus be understood as an endogenous weighted attention network stemming from rational, cognitive cost-constrained reasoning. Even under overload, agents reduce overall responsiveness while continuing to assign relatively higher weight to more reliable sources.

Neighbor Averaging as a Frictionless Limit

Let Ω denote the matrix of influence weights implied by the learning rule, with (i, j) entry $\Omega_{ij} := \omega_{ij}^*$ for $j \in N_i(g) \cup \{i\}$ and $\Omega_{ij} = 0$ otherwise. Under the quadratic cognitive cost, row sums satisfy

$$\sum_{j \in N_i(g) \cup \{i\}} \Omega_{ij} < 1 \quad \text{whenever } \lambda_i > 0,$$

reflecting anchoring on the prior via c_i^* and the cognitive deflator.

In the frictionless limit where cognitive costs disappear and the prior signal is uninformative, the model converges to DeGroot-style averaging:

Corollary 5. *Fix d_i and suppose (i) $\lambda_i \to 0$ and (ii) $\sigma^2_\theta \to \infty$. Then $c_i^* \to 0$ and*

$$\sum_{j \in N_i(g) \cup \{i\}} \Omega_{ij} \to 1.$$

Moreover, in the homoskedastic benchmark, $\Omega_{ij} \to 1/(d_i + 1)$ *for all* $j \in N_i(g) \cup \{i\}$*.*

Therefore, the benchmark DeGroot-style neighbor averaging emerges as a limiting case of our framework in which both cognitive costs and prior anchoring vanish.

7 Concluding Remarks

This paper addresses the tension between rational choice and the empirical observation that additional information can reduce decision quality. We develop a model in which information overload arises from rational learning under cognitive costs. In our framework, overload emerges endogenously from the trade-off between statistical accuracy and cognitive burden.

Our analysis delivers three main insights. First, we characterize conditions under which learning collapses in high-dimensional environments. When the marginal cost of responsiveness exceeds the marginal value of precision, agents attenuate responsiveness and revert toward prior-based rules, generating the empirically documented inverted-U relationship between information and accuracy. Second, we show that under variance-monotone costs, the optimal learning rule is affine and includes a cognitive deflator that uniformly shrinks responsiveness as dimensionality increases. Third, the framework yields implications for behavior and networks. Interpreting the informational dimension as network degree, agents face a trade-off between accuracy and cognitive costs, leading agents to optimally choose degrees in the interior generating skewed network structures and correlation-neglect–like behavior through optimal shrinkage.

These results imply that the value of additional information depends not only on its statistical content, but also on the dimensionality of the information environment. Policies that expand information without reducing processing costs may therefore reduce decision quality. In fact, even well-intentioned policies like those aimed at increasing transparency can lower the quality of decisions. This suggests a role for information design that emphasizes simplification and dimensionality reduction, for example, by limiting options or structuring information in specific ways.

These policy implications are closely related to the findings of Mani et al. (2013), who show that poverty itself taxes cognitive resources and impairs performance. While they focus on how scarcity and financial stress reduce available cognitive capacity, our analysis highlights a complementary channel: policy environments that expand informational dimensionality can induce optimal disengagement and prior reversion. Together, the two perspectives imply that many policies aimed at low-income populations, such as welfare eligibility rules, financial disclosures, or compliance requirements, may impose cognitive burdens beyond those imposed by poverty itself. As a result, low take-up, inertia, or apparent mistakes may reflect rational responses to excessive informational complexity rather than lack of information or motivation, suggesting that effective policy design should prioritize simplification, timing, and dimensionality reduction over additional disclosure or choice expansion.

Finally, our results have broader implications for information design. When expanding the set of available signals can optimally induce a collapse in learning, effective policymaking may require limiting, filtering, or structuring information rather than increasing disclosure indiscriminately. Against this backdrop, our framework also suggests natural extensions to network settings. Interpreting informational dimension as degree connectivity, the results point to models of endogenous network formation in which agents optimally choose degree under cognitive constraints, generating interior connectivity choices and skewed network structures.

Proofs

Proof of Lemma 1

Expand the mean squared error using the standard bias-variance decomposition:

$$E[(\ell-\theta)^2] \;=\; \mathrm{Var}(\theta)+\mathrm{Var}(\ell)+(E[\ell]-E[\theta])^2-2\,\mathrm{Cov}(\ell,\theta).$$

Since $(E[\ell]-E[\theta])^2 \geq 0$, we can drop the squared bias term. Applying the Cauchy–Schwarz inequality, $\mathrm{Cov}(\ell,\theta) \leq \sqrt{\mathrm{Var}(\ell)\,\mathrm{Var}(\theta)}$, we obtain the lower bound

$$E[(\ell-\theta)^2] \;\geq\; \mathrm{Var}(\theta)+\mathrm{Var}(\ell)-2\sqrt{\mathrm{Var}(\ell)\,\mathrm{Var}(\theta)}.$$

Now, assume the condition $E[(\ell-\theta)^2] \leq \mathrm{Var}(\theta)-\varepsilon$. Substituting this into the inequality above yields

$$\mathrm{Var}(\theta)-\varepsilon \;\geq\; \mathrm{Var}(\theta)+\mathrm{Var}(\ell)-2\sqrt{\mathrm{Var}(\ell)\,\mathrm{Var}(\theta)}.$$

Subtracting $\mathrm{Var}(\theta)$ from both sides and rearranging terms implies

$$2\sqrt{\mathrm{Var}(\ell)\,\mathrm{Var}(\theta)} \;\geq\; \mathrm{Var}(\ell)+\varepsilon \;\geq\; \varepsilon.$$

Squaring both sides and dividing by $4\,\mathrm{Var}(\theta)$ delivers the result:

$$\mathrm{Var}(\ell) \;\geq\; \frac{\varepsilon^2}{4\,\mathrm{Var}(\theta)}.$$

Proof of Proposition 1

By Assumption 1, the constant rule $\ell(S^{(d)}) \equiv \mu_\theta$ is feasible and satisfies $C(\ell; d) = 0$ and $E[(\mu_\theta - \theta)^2] = \mathrm{Var}(\theta)$. Hence the optimal objective value is weakly bounded above by $\mathrm{Var}(\theta)$, and therefore

$$\mathrm{MSE}(d) + C(\ell_d^*; d) \le \mathrm{Var}(\theta) \qquad \text{for all } d \ge 0. \tag{7}$$

Since $C(\ell; d) \ge 0$ for all (ℓ, d),

$$\mathrm{MSE}(d) \;\le\; \mathrm{Var}(\theta) \qquad \text{for all } d \ge 0. \tag{8}$$

We prove $\mathrm{MSE}(d) \to \mathrm{Var}(\theta)$. Fix any $\varepsilon > 0$ and suppose, toward a contradiction, that there exist infinitely many d such that

$$\mathrm{MSE}(d) \le \mathrm{Var}(\theta) - \varepsilon. \tag{9}$$

Let $\{d_k\}_{k\ge1}$ be a strictly increasing sequence of indices satisfying (9). For each k, define the optimal estimator random variable

$$X_{d_k} \;:=\; \ell^*_{d_k}(S^{(d_k)}).$$

By Lemma 1, (9) implies a uniform lower bound on dispersion:

$$\mathrm{Var}(X_{d_k}) \ge \underline{v} \quad \text{for all } k, \qquad \text{where } \underline{v} := \frac{\varepsilon^2}{4\,\mathrm{Var}(\theta)} > 0. \tag{10}$$

From (7) and the fact that $\mathrm{MSE}(d_k) \ge 0$, we obtain

$$C(\ell^*_{d_k}; d_k) \le \mathrm{Var}(\theta) \qquad \text{for all } k. \tag{11}$$

For each $d \ge d_1$, let

$$k(d) := \max\{k : \; d_k \le d\}.$$

Let $\pi_{d_{k(d)},d} : \mathbb{R}^{d+1} \to \mathbb{R}^{d_{k(d)}+1}$ denote the coordinate projection $\pi_{d_{k(d)},d}(s_0, \dots, s_d) = (s_0, \dots, s_{d_{k(d)}})$. Define $\hat{\ell}_d \in \mathcal{L}^{(d)}$ by lifting $\ell^*_{d_{k(d)}}$ to environment d and ignoring additional coordinates:

$$\hat{\ell}_d(s_0, \dots, s_d) \;:=\; \ell^*_{d_{k(d)}}\big(\pi_{d_{k(d)},d}(s_0, \dots, s_d)\big) \;=\; \ell^*_{d_{k(d)}}(s_0, \dots, s_{d_{k(d)}}).$$

This rule is feasible by nesting/free disposal.

By the consistency, the joint distribution of $(\theta, \pi_{d_{k(d)},d}(S^{(d)}))$ coincides with the joint distribution of $(\theta, S^{(d_{k(d)})})$. Consequently, the joint distribution of $(\theta, \hat{\ell}_d(S^{(d)}))$ coincides with that of $(\theta, \ell^*_{d_{k(d)}}(S^{(d_{k(d)})}))$. In particular,

$$\mathrm{Var}(\hat{\ell}_d(S^{(d)})) \;=\; \mathrm{Var}\big(\ell^*_{d_{k(d)}}(S^{(d_{k(d)})})\big) \;\geq\; \underline{v} \qquad \text{for all } d \geq d_1,$$

where the inequality uses (10). Therefore,

$$\liminf_{d\to\infty} \mathrm{Var}(\hat{\ell}_d(S^{(d)})) \geq \underline{v} > 0. \tag{12}$$

Along the subsequence $d = d_k$, we have $k(d_k) = k$ and hence $\hat{\ell}_{d_k} = \ell^*_{d_k}$ by construction. Thus, by (11),

$$C(\hat{\ell}_{d_k}; d_k) = C(\ell^*_{d_k}; d_k) \leq \mathrm{Var}(\theta) \qquad \text{for all } k.$$

Hence $C(\hat{\ell}_d; d) \not\to +\infty$ as $d \to \infty$, since it is bounded along the diverging subsequence $\{d_k\}$. Together with (12), this contradicts Assumption 2, which requires that any sequence of rules with $\liminf_{d\to\infty} \mathrm{Var}(\ell_d(S^{(d)})) > 0$ must satisfy $C(\ell_d; d) \to +\infty$.

The contradiction shows that (9) can hold for only finitely many d. Thus for every $\varepsilon > 0$, there exists $D(\varepsilon)$ such that for all $d \geq D(\varepsilon)$,

$$\mathrm{MSE}(d) > \mathrm{Var}(\theta) - \varepsilon,$$

so $\liminf_{d\to\infty} \mathrm{MSE}(d) \geq \mathrm{Var}(\theta)$. Combining with the upper bound (8) yields

$$\lim_{d\to\infty} \mathrm{MSE}(d) = \mathrm{Var}(\theta).$$

■

Proof of Proposition 2

Necessity. If overload occurs, there exists d_1 such that $\mathrm{MSE}(d) > \mathrm{MSE}(d_1)$ for all $d > d_1$. By (1), $\mathrm{MSE}(d) \leq \mathrm{Var}(\theta)$ for all d. Hence it must be that $\mathrm{MSE}(d_1) < \mathrm{Var}(\theta)$, establishing (i) with $d_0 = d_1$. Condition (ii) follows from Proposition 1 (which requires only Assumptions 1–2).

Sufficiency. Suppose (i) and (ii) hold. Let d_0 satisfy $\mathrm{MSE}(d_0) < \mathrm{Var}(\theta)$ and define

$$\delta := \frac{\mathrm{Var}(\theta) - \mathrm{MSE}(d_0)}{2} \;>\; 0.$$

By (ii), there exists D such that for all $d \geq D$,

$$\mathrm{MSE}(d) \ > \ \mathrm{Var}(\theta) - \delta \ = \ \frac{\mathrm{Var}(\theta) + \mathrm{MSE}(d_0)}{2} \ > \ \mathrm{MSE}(d_0).$$

Let d_1 be the *largest* minimizer of $\mathrm{MSE}(d)$ on the finite set $\{0, 1, \ldots, D\}$:

$$d_1 \in \arg\min_{0 \leq d \leq D} \mathrm{MSE}(d), \quad \text{chosen so that } d_1 \text{ is maximal among minimizers.}$$

Then for any d with $d_1 < d \leq D$, maximality implies $\mathrm{MSE}(d) > \mathrm{MSE}(d_1)$; and for any $d > D$, the bound above implies $\mathrm{MSE}(d) > \mathrm{MSE}(d_0) \geq \mathrm{MSE}(d_1)$, hence also $\mathrm{MSE}(d) > \mathrm{MSE}(d_1)$. Therefore $\mathrm{MSE}(d) > \mathrm{MSE}(d_1)$ for all $d > d_1$, which is overload.

■

Proof of Proposition 3

Properness and lower semicontinuity. By Assumption 3(i), $C(\cdot; d)$ is finite somewhere, and $E[(\ell - \theta)^2] < \infty$ for all $\ell \in \mathcal{L}^{(d)}$ by definition. Hence J_d is proper. The map $\ell \mapsto E[(\ell - \theta)^2]$ is continuous in $\| \cdot \|_2$, and by Assumption 3(iii), $C(\cdot; d)$ is lower semicontinuous, so J_d is lower semicontinuous.

Coercivity. Expand

$$E[(\ell - \theta)^2] = E[\ell^2] - 2E[\ell\theta] + E[\theta^2] \geq \|\ell\|_2^2 - 2\|\ell\|_2\|\theta\|_2 + E[\theta^2],$$

where the inequality is Cauchy–Schwarz: $E[\ell\theta] \leq \|\ell\|_2\|\theta\|_2$. Thus $E[(\ell - \theta)^2] \to \infty$ as $\|\ell\|_2 \to \infty$. Since $C(\ell; d) \geq 0$, also $J_d(\ell) \to \infty$ as $\|\ell\|_2 \to \infty$. Hence J_d is coercive.

Existence. A proper, coercive, lower semicontinuous convex functional on a Hilbert space attains its minimum on any closed convex domain. Since $\mathcal{L}^{(d)}$ is closed in L^2, J_d attains a minimum on $\mathcal{L}^{(d)}$.

Uniqueness. Let $\ell_1 \neq \ell_2$ with $Pr(\ell_1 \neq \ell_2) > 0$ and fix $\alpha \in (0, 1)$. Then

$$E\big[(\alpha\ell_1 + (1 - \alpha)\ell_2 - \theta)^2\big] < \alpha E[(\ell_1 - \theta)^2] + (1 - \alpha)E[(\ell_2 - \theta)^2],$$

because the map $\ell \mapsto E[(\ell - \theta)^2]$ is *strictly* convex on L^2. By convexity of $C(\cdot; d)$,

$$C(\alpha\ell_1 + (1 - \alpha)\ell_2; d) \leq \alpha C(\ell_1; d) + (1 - \alpha)C(\ell_2; d).$$

Adding these two inequalities yields strict convexity of J_d, so J_d cannot have two distinct minimizers. Hence the minimizer is unique up to a.s. equality.

■

Proof of Proposition 4

Fix d and work in L^2 with inner product $\langle X,Y\rangle := E[XY]$. Let $\mathcal{A}^{(d)} := \{\omega^\top S^{(d)} + c : (\omega,c) \in \mathbb{R}^{d+1} \times \mathbb{R}\}$. This is a finite-dimensional (hence closed) linear subspace of $\mathcal{L}^{(d)}$. Let $P_{\mathcal{A}^{(d)}} : \mathcal{L}^{(d)} \to \mathcal{A}^{(d)}$ denote the orthogonal projection.

Take any feasible learning rule $\ell \in \mathcal{L}^{(d)}$ and decompose it as

$$\ell = a + r, \qquad a := P_{\mathcal{A}^{(d)}}(\ell) \in \mathcal{A}^{(d)}, \qquad r := \ell - a \in (\mathcal{A}^{(d)})^\perp.$$

Then $\langle r,h\rangle = 0$ for all $h \in \mathcal{A}^{(d)}$.

Let $m := E[\theta \mid S^{(d)}]$ and write $\theta = m + e$ where $E[e \mid S^{(d)}] = 0$. By Assumption 4, we have $m \in \mathcal{A}^{(d)}$. Since $\ell - m$ is measurable with respect to $S^{(d)}$,

$$E[(\ell - m)e] = E\big[E[(\ell - m)e \mid S^{(d)}]\big] = E[(\ell - m)E[e \mid S^{(d)}]] = 0.$$

Therefore,

$$E[(\ell - \theta)^2] = E[(\ell - m)^2] + E[e^2]. \tag{13}$$

Next note that $\ell - m = (a - m) + r$ with $(a - m) \in \mathcal{A}^{(d)}$ and $r \perp \mathcal{A}^{(d)}$, hence

$$E[(\ell - m)^2] = E[(a - m)^2] + E[r^2].$$

Combining this with (13) and observing that $E[(a-\theta)^2] = E[(a-m)^2] + E[e^2]$, we obtain

$$E[(\ell - \theta)^2] = E[(a - \theta)^2] + E[r^2] \;\geq\; E[(a - \theta)^2], \tag{14}$$

with strict inequality whenever $P(r \neq 0) > 0$.

Because constants belong to $\mathcal{A}^{(d)}$ (take $\omega = 0$), we have $1 \in \mathcal{A}^{(d)}$. Since $r \perp 1$, it follows that $E[r] = 0$. Moreover,

$$\mathrm{Cov}(a,r) = E[(a - E[a])r] = \langle a - E[a],\, r\rangle = 0,$$

because $a - E[a] \in \mathcal{A}^{(d)}$ and $r \perp \mathcal{A}^{(d)}$. Hence

$$\mathrm{Var}(\ell) = \mathrm{Var}(a+r) = \mathrm{Var}(a) + \mathrm{Var}(r) + 2\,\mathrm{Cov}(a,r) = \mathrm{Var}(a) + \mathrm{Var}(r) \;\geq\; \mathrm{Var}(a). \tag{15}$$

By Assumption 5 and (15),

$$C(\ell; d) \geq C(a; d).$$

Combining (14) with cost monotonicity yields

$$J_d(\ell) = E[(\ell - \theta)^2] + C(\ell; d) \;\geq\; E[(a - \theta)^2] + C(a; d) = J_d(a).$$

Thus every feasible ℓ is weakly dominated by its affine projection $a \in \mathcal{A}^{(d)}$.

Finally, let ℓ_d^* be the unique minimizer of (P_d) from Proposition 3. If $\ell_d^* \notin \mathcal{A}^{(d)}$, then its residual r is nonzero with positive probability, so (14) implies $J_d(P_{\mathcal{A}^{(d)}}(\ell_d^*)) < J_d(\ell_d^*)$, contradicting optimality. Therefore $\ell_d^* \in \mathcal{A}^{(d)}$, i.e., $\ell_d^*(S^{(d)}) = \omega_d^{*\top} S^{(d)} + c_d^*$ for some (ω_d^*, c_d^*).

■

Proof of Proposition 5

Fix $d \geq 0$ and write $m := d+1$. Under Assumption 6, the signal vector satisfies $S^{(d)} = \theta \mathbf{1}_m + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2 I_m)$ is independent of θ. Under the quadratic processing cost,

$$C_r(\ell; d) = \lambda m \operatorname{Var}(\ell(S^{(d)})).$$

By Proposition 4 (Affine Optimality), we can restrict attention to affine rules $\ell(S^{(d)}) = \omega^\top S^{(d)} + c$.

For any fixed ω, the objective in (P_d) is

$$E\big[(\omega^\top S^{(d)} + c - \theta)^2\big] + \lambda m \operatorname{Var}(\omega^\top S^{(d)} + c).$$

The variance term does not depend on c, so the minimizing c is the L^2 projection of $\theta - \omega^\top S^{(d)}$ onto constants:

$$0 = \frac{\partial}{\partial c} E\big[(\omega^\top S^{(d)} + c - \theta)^2\big] = 2\, E\big[\omega^\top S^{(d)} + c - \theta\big],$$

hence

$$c = \mu_\theta - \omega^\top E[S^{(d)}].$$

Since $E[S^{(d)}] = \mu_\theta \mathbf{1}_m$, this yields

$$c = \big(1 - \omega^\top \mathbf{1}_m\big)\mu_\theta.$$

This is exactly the expression in the proposition once we identify $\omega = \omega_d^*$.

Define centered variables $\tilde{\theta} := \theta - \mu_\theta$ and $\tilde{S} := S^{(d)} - \mu_\theta \mathbf{1}_m$. With the optimal c from above, we have

$$\ell(S^{(d)}) - \mu_\theta = \omega^\top \tilde{S}, \qquad \ell(S^{(d)}) - \theta = \omega^\top \tilde{S} - \tilde{\theta}.$$

Let $\Sigma := \mathrm{Var}(\tilde{S}) = \mathrm{Var}(S^{(d)})$. Under Assumption 6,

$$\Sigma = \sigma_\theta^2 \mathbf{1}_m \mathbf{1}_m^\top + \sigma_\varepsilon^2 I_m, \qquad \mathrm{Cov}(\tilde{S}, \tilde{\theta}) = \sigma_\theta^2 \mathbf{1}_m.$$

Expanding the objective yields

$$\begin{aligned} E\big[(\omega^\top \tilde{S} - \tilde{\theta})^2\big] &= E[(\omega^\top \tilde{S})^2] - 2E[\omega^\top \tilde{S}\,\tilde{\theta}] + E[\tilde{\theta}^2] \\ &= \omega^\top \Sigma \omega - 2\omega^\top (\sigma_\theta^2 \mathbf{1}_m) + \sigma_\theta^2, \end{aligned}$$

and

$$\mathrm{Var}(\ell(S^{(d)})) = \mathrm{Var}(\omega^\top \tilde{S}) = \omega^\top \Sigma \omega.$$

Therefore the problem reduces to minimizing over $\omega \in \mathbb{R}^m$:

$$\sigma_\theta^2 - 2\sigma_\theta^2\, \omega^\top \mathbf{1}_m + (1 + \lambda m)\, \omega^\top \Sigma \omega.$$

This is a strictly convex quadratic (since $\Sigma \succ 0$ and $1 + \lambda m > 0$), hence the minimizer is unique and satisfies the first-order condition

$$-2\sigma_\theta^2 \mathbf{1}_m + 2(1 + \lambda m)\Sigma\omega = 0, \quad \text{so} \quad \omega = \frac{\sigma_\theta^2}{1 + \lambda m}\, \Sigma^{-1} \mathbf{1}_m.$$

Write $\Sigma = \sigma_\varepsilon^2 I_m + \sigma_\theta^2 \mathbf{1}_m \mathbf{1}_m^\top$. By the Sherman–Morrison formula,

$$(\sigma_\varepsilon^2 I_m + \sigma_\theta^2 \mathbf{1}\mathbf{1}^\top)^{-1} = \frac{1}{\sigma_\varepsilon^2} I_m - \frac{\sigma_\theta^2}{\sigma_\varepsilon^2\big(\sigma_\varepsilon^2 + \sigma_\theta^2 m\big)}\, \mathbf{1}\mathbf{1}^\top,$$

hence

$$\Sigma^{-1}\mathbf{1}_m = \left(\frac{1}{\sigma_\varepsilon^2} - \frac{\sigma_\theta^2 m}{\sigma_\varepsilon^2(\sigma_\varepsilon^2 + \sigma_\theta^2 m)}\right)\mathbf{1}_m = \frac{1}{\sigma_\varepsilon^2 + \sigma_\theta^2 m}\, \mathbf{1}_m.$$

Substituting into $\omega = \frac{\sigma_\theta^2}{1+\lambda m}\Sigma^{-1}\mathbf{1}_m$ gives

$$\omega_d^* = \alpha_d\, \mathbf{1}_m, \qquad \alpha_d = \frac{1}{1 + \lambda(d+1)} \cdot \frac{\sigma_\theta^2}{\sigma_\varepsilon^2 + \sigma_\theta^2 (d+1)}.$$

Finally, using the optimal c yields

$$c_d^* = \big(1 - (\omega_d^*)^\top \mathbf{1}_m\big)\mu_\theta.$$

■

Proof of Proposition 6

Maintain Assumption 6 and the quadratic cost. For each $d \geq 0$, define the minimized *total loss*

$$L(d) := \min_{\ell \in \mathcal{L}^{(d)}} \left\{ E\big[(\ell(S^{(d)}) - \theta)^2\big] + \lambda(d+1)\,\mathrm{Var}(\ell(S^{(d)}))\right\}.$$

By Proposition 5, the minimizer is affine with coefficient vector ω_d^*, and the reduced problem in ω is a strictly convex quadratic. A standard quadratic-completion argument yields the minimized value

$$L(d) = \sigma_\theta^2 - \frac{\sigma_\theta^4}{1+\lambda(d+1)}\,\mathbf{1}_m^\top \Sigma^{-1}\mathbf{1}_m, \qquad m = d+1, \qquad \Sigma = \sigma_\varepsilon^2 I_m + \sigma_\theta^2 \mathbf{1}_m \mathbf{1}_m^\top.$$

From the computation in the proof of Proposition 5, $\Sigma^{-1}\mathbf{1}_m = \frac{1}{\sigma_\varepsilon^2 + \sigma_\theta^2 m}\mathbf{1}_m$, hence

$$\mathbf{1}_m^\top \Sigma^{-1}\mathbf{1}_m = \frac{m}{\sigma_\varepsilon^2 + \sigma_\theta^2 m}.$$

Therefore,

$$L(d) = \sigma_\theta^2 - \frac{\sigma_\theta^4 (d+1)}{(1+\lambda(d+1))\big(\sigma_\varepsilon^2 + \sigma_\theta^2(d+1)\big)}.$$

Let $m = d+1 > 0$ be real valued and define

$$\widetilde{L}(m) := \sigma_\theta^2 - \frac{\sigma_\theta^4 m}{(1+\lambda m)(\sigma_\varepsilon^2 + \sigma_\theta^2 m)}.$$

Differentiating yields

$$\widetilde{L}'(m) = -\frac{\sigma_\theta^4\big(\sigma_\varepsilon^2 - \lambda\sigma_\theta^2 m^2\big)}{(\sigma_\varepsilon^2 + \sigma_\theta^2 m)^2 (1+\lambda m)^2},$$

so $\widetilde{L}'(m) < 0$ when $m < \sqrt{\sigma_\varepsilon^2/(\lambda\sigma_\theta^2)}$ and $\widetilde{L}'(m) > 0$ when $m > \sqrt{\sigma_\varepsilon^2/(\lambda\sigma_\theta^2)}$ (for $\lambda > 0$). Hence $\widetilde{L}$ is strictly decreasing up to the unique minimizer

$$m^* := \sqrt{\frac{\sigma_\varepsilon^2}{\lambda\sigma_\theta^2}},$$

and strictly increasing thereafter. (When $\lambda = 0$, $\widetilde{L}$ is strictly decreasing in m, implying it admits a corner solution.)

Because $\widetilde{L}$ is decreasing then increasing, the integer sequence $L(d)$ is unimodal: it strictly decreases while $d+1 < m^*$ and strictly increases once $d+1 > m^*$. Thus the only remaining question is whether the sequence is already increasing at $d = 0$.

Compute the first increment:

$$L(1) - L(0) = \widetilde{L}(2) - \widetilde{L}(1).$$

Using the closed form above, we have

$$\widetilde{L}(2) - \widetilde{L}(1) = -\frac{\sigma_\theta^4\big(\sigma_\varepsilon^2 - 2\lambda\sigma_\theta^2\big)}{(1+2\lambda)(\sigma_\varepsilon^2 + 2\sigma_\theta^2)(1+\lambda)(\sigma_\varepsilon^2 + \sigma_\theta^2)}.$$

The denominator is strictly positive, so the sign is determined by $\sigma_\varepsilon^2 - 2\lambda\sigma_\theta^2$.

Case (i): $\lambda > \sigma_\varepsilon^2/(2\sigma_\theta^2)$. Then $\widetilde{L}(2) > \widetilde{L}(1)$, i.e. $L(1) > L(0)$. Moreover, $\lambda > \sigma_\varepsilon^2/(2\sigma_\theta^2)$ implies $m^* < \sqrt{2} < 2$. Since $\widetilde{L}$ is strictly increasing for all $m \geq 2$, it follows that $L(d)$ is strictly increasing for every integer $d \geq 0$.

Case (ii): $\lambda \leq \sigma_\varepsilon^2/(2\sigma_\theta^2)$. Then $\widetilde{L}(2) \leq \widetilde{L}(1)$, i.e. $L(1) \leq L(0)$, so the sequence does not start in the increasing region. Because $\widetilde{L}$ is unimodal and tends to σ_θ^2 as $m \to \infty$, there exists at least one minimizer over integers; unimodality implies that there is an integer $d^* \geq 0$ such that $L(d)$ strictly decreases for $d < d^*$ and strictly increases for $d > d^*$ (with possible tie only in the knife-edge equality case).

■

Proof of Proposition 7

By Proposition 4, the optimal rule is affine: $\ell(S^{(d)}) = \omega^\top S^{(d)} + c$. Indeed, under Assumption 7, the vector $(\theta, S^{(d)})$ is jointly Gaussian, so the conditional expectation is affine, and with quadratic cognitive costs the induced cost is variance-monotone and satisfies Assumption 3.

As in the proof of Proposition 5, the optimal intercept satisfies $c = \mu_\theta - \omega^\top E[S^{(d)}]$ and therefore $c = \big(1 - \omega^\top \mathbf{1}_m\big)\mu_\theta$. Let $\tilde{S} = S^{(d)} - \mu_\theta \mathbf{1}_m$ and $\tilde{\theta} = \theta - \mu_\theta$. Then the objective becomes

$$\sigma_\theta^2 - 2\sigma_\theta^2\, \omega^\top \mathbf{1}_m + (1 + \lambda m)\, \omega^\top \Sigma\, \omega,$$

where now

$$\Sigma = \mathrm{Var}(S^{(d)}) = \sigma_\theta^2 \mathbf{1}_m \mathbf{1}_m^\top + D, \qquad D := diag(\sigma_{\varepsilon,0}^2, \dots, \sigma_{\varepsilon,d}^2).$$

Strict convexity implies the unique minimizer satisfies

$$(1+\lambda m)\Sigma\omega = \sigma_\theta^2 \mathbf{1}_m, \qquad \text{so} \qquad \omega = \frac{\sigma_\theta^2}{1+\lambda m}\,\Sigma^{-1}\mathbf{1}_m.$$

To compute $\Sigma^{-1}\mathbf{1}_m$, apply the Woodbury identity with $\Sigma = D + \sigma_\theta^2 \mathbf{1}\mathbf{1}^\top$:

$$\Sigma^{-1} = D^{-1} - D^{-1}\mathbf{1}\left(\frac{1}{\sigma_\theta^2} + \mathbf{1}^\top D^{-1}\mathbf{1}\right)^{-1} \mathbf{1}^\top D^{-1}.$$

Let $P_d := \mathbf{1}^\top D^{-1}\mathbf{1} = \sum_{k=0}^{d} \sigma_{\varepsilon,k}^{-2}$. Then

$$\begin{aligned}\Sigma^{-1}\mathbf{1} &= D^{-1}\mathbf{1} - D^{-1}\mathbf{1}\left(\frac{1}{\sigma_\theta^2} + P_d\right)^{-1} \mathbf{1}^\top D^{-1}\mathbf{1} \\ &= D^{-1}\mathbf{1}\left(1 - \frac{P_d}{\frac{1}{\sigma_\theta^2} + P_d}\right) = \frac{1}{1+\sigma_\theta^2 P_d}\, D^{-1}\mathbf{1}.\end{aligned}$$

Thus

$$\omega = \frac{\sigma_\theta^2}{1+\lambda m} \cdot \frac{1}{1+\sigma_\theta^2 P_d}\, D^{-1}\mathbf{1},$$

so each component satisfies

$$\omega_{d,k}^* = \frac{1}{1+\lambda(d+1)} \cdot \frac{\sigma_\theta^2}{1+\sigma_\theta^2 P_d} \cdot \frac{1}{\sigma_{\varepsilon,k}^2}, \qquad k = 0, 1, \dots, d.$$

The monotonicity statement $\sigma_{\varepsilon,k}^2 < \sigma_{\varepsilon,j}^2 \Leftrightarrow \omega_{d,k}^* > \omega_{d,j}^*$ is immediate from the last display.

■

Proof of Proposition 8

Fix $\rho \in (0,1)$. Let $m = d+1$. By Remark 1, the optimal (correlation-aware) coefficient vector is $\omega^* = \alpha^* \mathbf{1}_m$ with

$$\alpha^* = \frac{\sigma_\theta^2}{(1+\lambda m)\big(\sigma_\varepsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\varepsilon^2 \rho)\big)}.$$

The correlation-neglect rule treats the signals as independent (i.e. uses $\rho = 0$ in the same optimization), so it has $\omega^N = \alpha^N \mathbf{1}_m$ with

$$\alpha^N = \frac{\sigma_\theta^2}{(1+\lambda m)\big(\sigma_\varepsilon^2 + m\sigma_\theta^2\big)}.$$

Therefore,

$$\|\omega^* - \omega^N\|_2 = \sqrt{m}\,|\alpha^* - \alpha^N|.$$

Compute the difference:

$$\begin{aligned}
\alpha^* - \alpha^N &= \frac{\sigma_\theta^2}{1+\lambda m}\left(\frac{1}{\sigma_\varepsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\varepsilon^2\rho)} - \frac{1}{\sigma_\varepsilon^2 + m\sigma_\theta^2}\right) \\
&= \frac{\sigma_\theta^2}{1+\lambda m} \cdot \frac{\big(\sigma_\varepsilon^2 + m\sigma_\theta^2\big) - \big(\sigma_\varepsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\varepsilon^2\rho)\big)}{\big(\sigma_\varepsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\varepsilon^2\rho)\big)\big(\sigma_\varepsilon^2 + m\sigma_\theta^2\big)} \\
&= \frac{\sigma_\theta^2}{1+\lambda m} \cdot \frac{\sigma_\varepsilon^2\rho(1-m)}{\big(\sigma_\varepsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\varepsilon^2\rho)\big)\big(\sigma_\varepsilon^2 + m\sigma_\theta^2\big)}.
\end{aligned}$$

Taking absolute values and using $\rho \in [0, 1)$,

$$|\alpha^* - \alpha^N| = \frac{\sigma_\theta^2\sigma_\varepsilon^2\rho(m-1)}{(1+\lambda m)\big(\sigma_\varepsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\varepsilon^2\rho)\big)\big(\sigma_\varepsilon^2 + m\sigma_\theta^2\big)}.$$

Since $\sigma_\varepsilon^2 + m\sigma_\theta^2 \geq m\sigma_\theta^2$ and $\sigma_\varepsilon^2(1-\rho) + m(\sigma_\theta^2 + \sigma_\varepsilon^2\rho) \geq m\sigma_\theta^2$ for $\rho \geq 0$, we obtain the bound

$$|\alpha^* - \alpha^N| \leq \frac{\sigma_\theta^2\sigma_\varepsilon^2\rho(m-1)}{(1+\lambda m)\,(m\sigma_\theta^2)\,(m\sigma_\theta^2)} = \frac{\sigma_\varepsilon^2\rho}{1+\lambda m} \cdot \frac{m-1}{m^2} \cdot \frac{1}{\sigma_\theta^2}.$$

Multiplying by $\sqrt{m}$ gives

$$\|\omega^* - \omega^N\|_2 \leq \frac{\sigma_\varepsilon^2\rho}{\sigma_\theta^2} \cdot \frac{1}{1+\lambda m} \cdot \frac{\sqrt{m}(m-1)}{m^2} = O\!\left(\frac{1}{(1+\lambda(d+1))(d+1)^{1/2}}\right).$$

■

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